Practical 2c: Constructor Overloading

AIM : To demonstrate constructor overloading in C# by defining multiple constructors with different parameter lists in the same class.

PROGRAM CODE

```
using System;

namespace ConstructorOverloading
{
   class Student
   {
      public string Name;
      public int Age;
      public string Course;

      // Default constructor
      public Student()
      {
         Name = "Unknown";
         Age = 0;
         Course = "N/A";
         Console.WriteLine("Default constructor called");
      }

      // Parameterized constructor with 1 argument
      public Student(string name)
      {
         Name = name;
         Age = 18;
         Course = "BSc IT";
         Console.WriteLine("1-param constructor called");
      }

      // Parameterized constructor with 2 arguments
      public Student(string name, int age)
      {
         Name = name;
```

```
        Age = age;
        Course = "BSc IT";
        Console.WriteLine("2-param constructor called");
    }

    // Parameterized constructor with all arguments
    public Student(string name, int age, string course)
    {
        Name = name;
        Age = age;
        Course = course;
        Console.WriteLine("3-param constructor called");
    }

    public void Display()
    {
        Console.WriteLine($"Name: {Name}, Age: {Age}, Course: {Course}");
    }
}

class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("=== Constructor Overloading Demo ===");
        Student s1 = new Student();
        s1.Display();

        Student s2 = new Student("Alice");
        s2.Display();

        Student s3 = new Student("Bob", 20);
        s3.Display();

        Student s4 = new Student("Carol", 21, "BCA");
        s4.Display();

        Console.ReadKey();
    }
}
```

}

OUTPUT

```
=== Constructor Overloading Demo ===
Default constructor called
Name: Unknown, Age: 0, Course: N/A
1-param constructor called
Name: Alice, Age: 18, Course: BSc IT
2-param constructor called
Name: Bob, Age: 20, Course: BSc IT
3-param constructor called
Name: Carol, Age: 21, Course: BCA
```